

Spatially continuous (raster) covariates by intensity-scale aggregation

Why averaging predictors over polygons is biased, and the `SDALGCP2_raster` fix

SDALGCP2 development notes

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Abstract

When a covariate varies *within* an areal unit it is usually summarised by its areal mean and entered into a Poisson model. Under the nonlinear log link this is biased. We derive the statistically coherent alternative used by `SDALGCP2_raster`: covariates supplied as rasters enter the log-Gaussian Cox process intensity at the candidate-point level and are aggregated on the *intensity* (exponential) scale through a log-sum-exp offset. We give the exact discretised model, the factorisation that isolates the covariate contribution, the Gauss–Newton fixed point used to fit it (which reuses the ordinary SDA-LGCP machinery), and a simulation in which areal averaging is biased by +67% while the intensity-scale fit recovers the truth.

1 The aggregated LGCP mean

Let the LGCP intensity be $\lambda(x) = \rho(x) \exp\{z(x)^\top \beta + S(x)\}$, with $\rho(x)$ the population (at-risk) density, $z(x)$ a vector of covariates available as rasters, and S a zero-mean Gaussian process with $\text{Cov}\{S(x), S(x')\} = \sigma^2 \rho_\phi(\|x - x'\|)$. The count in region A_i is conditionally Poisson with mean $\int_{A_i} \lambda(x) dx$. Writing $m_i = \int_{A_i} \rho(x) dx$ for the population at risk and $w_i(x) = \rho(x)/m_i$ (so $\int_{A_i} w_i = 1$),

$$\mathbb{E}[Y_i | S] = m_i \int_{A_i} w_i(x) \exp\{z(x)^\top \beta + S(x)\} dx. \quad (1)$$

With the same candidate points x_{ik} and quadrature weights w_{ik} used to build the aggregated correlation (companion note on the scale parameter),

$$\mathbb{E}[Y_i | S] \approx m_i \sum_{k=1}^{n_i} w_{ik} \exp\{z(x_{ik})^\top \beta + S(x_{ik})\}. \quad (2)$$

2 Why averaging the predictor over polygons is biased

The conventional “areal” model replaces $z(x)$ by its within-region mean $\bar{z}_i = \sum_k w_{ik} z(x_{ik})$ and sets the region linear predictor to $\bar{z}_i^\top \beta$. But by Jensen’s inequality, for the covariate part alone,

$$\sum_k w_{ik} \exp\{z(x_{ik})^\top \beta\} \geq \exp\{\bar{z}_i^\top \beta\}, \quad (3)$$

with equality only if z is constant on A_i . The gap is governed by the *within-region* variance of $z^\top \beta$. Crucially, a regression of Y_i on $\bar{z}_i$ is biased *for the coefficient* β precisely when that within-region variance is correlated with $\bar{z}_i$ across regions — for example when the covariate has sharp

localised peaks (point exposure sources): regions containing a peak have a moderate mean but a large intensity aggregate. Averaging washes the peaks out; the exponential does not.

3 Intensity-scale factorisation

Factor the covariate out of (2):

$$\boxed{\mathbb{E}[Y_i | S] = m_i e^{b_i(\beta)} \sum_k c_{ik}(\beta) e^{S(x_{ik})}, \quad b_i(\beta) = \log \sum_k w_{ik} e^{z(x_{ik})^\top \beta}} \quad (4)$$

where the *intensity-tilted* weights

$$c_{ik}(\beta) = \frac{w_{ik} e^{z(x_{ik})^\top \beta}}{\sum_l w_{il} e^{z(x_{il})^\top \beta}}, \quad \sum_k c_{ik} = 1, \quad (5)$$

upweight high-intensity sub-areas. The term $b_i(\beta)$ is the **log-sum-exp** of the within-region covariate surface; it equals $\bar{z}_i^\top \beta$ only when z is constant on A_i , and is the correct region-level covariate contribution.

4 Reducing the spatial part

Let $T_i = \sum_k c_{ik} S(x_{ik})$, a Gaussian linear functional, and write $S(x_{ik}) = T_i + \delta_{ik}$ with $\sum_k c_{ik} \delta_{ik} = 0$. A second-order expansion gives

$$\sum_k c_{ik} e^{S(x_{ik})} = e^{T_i} \sum_k c_{ik} e^{\delta_{ik}} \approx e^{T_i} \left(1 + \frac{1}{2} \sum_k c_{ik} \delta_{ik}^2 \right) \approx \exp(T_i + \frac{1}{2} v_i), \quad (6)$$

with $v_i = \mathbb{E} \sum_k c_{ik} \delta_{ik}^2 = \sigma^2 (1 - \sum_{k,l} c_{ik} c_{il} \rho_\phi(d_{kl})) = \sigma^2 (1 - \mathbf{R}_{ii}^c)$, the variance lost to spatial averaging. Hence the *full* (covariate-tilted) model is

$$\mathbb{E}[Y_i | T_i] = m_i \exp(b_i(\beta) + \frac{1}{2} v_i(\beta) + T_i), \quad \mathbf{T} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{R}^c(\phi, \beta)), \quad \mathbf{R}_{ij}^c = \sum_{k,l} c_{ik} c_{jl} \rho_\phi(d_{kl}). \quad (7)$$

Offset-only model (the default implementation). Because the dominant correction is the covariate term b_i , the implemented default keeps the spatial aggregation at the population weights ($\mathbf{R}^c \rightarrow \mathbf{R}^w$, $\frac{1}{2} v_i$ absorbed), so the correlation matrix does *not* depend on β and is precomputed once:

$$\mathbb{E}[Y_i | T_i] = m_i \exp(b_i(\beta) + T_i), \quad \mathbf{T} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{R}^w(\phi)). \quad (8)$$

This captures the covariate Jensen bias *exactly* through $b_i(\beta)$ and treats the covariate tilting of the latent field as second order. The fully tilted model (7) is available as a refinement (it rebuilds $\mathbf{R}^c(\beta)$ each outer iteration).

5 Estimation: a Gauss–Newton fixed point

The only nonlinearity in β is through $b_i(\beta)$, whose gradient is the *intensity-tilted* (“effective”) covariate

$$\frac{\partial b_i}{\partial \beta} = \frac{\sum_k w_{ik} e^{z_{ik}^\top \beta} z_{ik}}{\sum_k w_{ik} e^{z_{ik}^\top \beta}} = \sum_k c_{ik}(\beta) z(x_{ik}) \equiv \mathbf{D}_i^c(\beta). \quad (9)$$

At iteration t we linearise $b_i(\boldsymbol{\beta}) \approx b_i(\boldsymbol{\beta}^{(t)}) + \mathbf{D}_i^c(\boldsymbol{\beta}^{(t)})^\top (\boldsymbol{\beta} - \boldsymbol{\beta}^{(t)})$ and fit the ordinary SDA-LGCP MCML with *effective design* $\mathbf{D}^c(\boldsymbol{\beta}^{(t)})$ and offset $o_i^{(t)} = \log m_i + b_i(\boldsymbol{\beta}^{(t)}) - \mathbf{D}_i^c(\boldsymbol{\beta}^{(t)})^\top \boldsymbol{\beta}^{(t)}$, obtaining $\boldsymbol{\beta}^{(t+1)}$ (and σ^2, ϕ). At a fixed point $\boldsymbol{\beta}^*$ the offset reproduces $b_i(\boldsymbol{\beta}^*)$ exactly and, by (9), the linear score $\sum_i \mathbf{D}_i^c(\cdot)$ equals the nonlinear score $\sum_i (\partial b_i / \partial \boldsymbol{\beta})(\cdot)$, so $\boldsymbol{\beta}^*$ solves the likelihood equations of the nonlinear model. The scheme therefore reuses `mcml_fit()` unchanged and is exactly Gauss–Newton on the $b_i(\boldsymbol{\beta})$ term. (`SDALGCP2_raster`; the iteration converges in a handful of steps.)

6 Numerical demonstration

A covariate with 14 sharp Gaussian peaks (width ~ 0.5 , region width 2.5) on an 8×8 lattice; counts simulated from (2) with $\beta_z = 1$.

method	$\widehat{\beta_z}$	bias
truth	1.00	—
naive (areal-mean covariate, Poisson GLM)	1.67	+67%
exact nonlinear MLE of (2)	1.01	+1%
<code>SDALGCP2_raster</code> (offset model)	1.06	+6%

Areal averaging is badly biased; intensity-scale aggregation recovers the truth and matches the exact MLE. The bias is large only when the within-region covariate variance correlates with the region mean (sharp sub-region features); for smooth covariates the two approaches nearly agree, as expected from $b_i(\boldsymbol{\beta}) \approx \bar{z}_i^\top \boldsymbol{\beta} + \frac{1}{2} \boldsymbol{\beta}^\top \widehat{\text{Var}}_i(z) \boldsymbol{\beta}$.

7 Spatio-temporal extension

When the same regions are observed at times $t = 1, \dots, T$, the count Y_{it} is conditionally Poisson with mean $m_{it} e^{\eta_{it}}$ and a separable space–time latent field

$$\eta_{it} = b_i(\boldsymbol{\beta}) + S_{it}, \quad \text{Cov}(\text{vec } \mathbf{S}) = \sigma^2 \mathbf{R}_t(\nu) \otimes \mathbf{R}_s(\phi), \quad (10)$$

with $\mathbf{R}_s(\phi)$ the aggregated spatial correlation of the previous sections and $\mathbf{R}_t(\nu)$ a temporal Matérn correlation (range ν , smoothness κ_t); the likelihood is evaluated without forming the $(NT) \times (NT)$ covariance (companion spatio-temporal note).

We take the covariate surface $z(x)$ to be *time-invariant* (genuinely time-varying covariates enter instead as ordinary region–time columns of the design). Then the log-sum-exp offset $b_i(\boldsymbol{\beta})$ and the intensity-tilted weights $c_{ik}(\boldsymbol{\beta})$ of (4)–(5) are *exactly the spatial ones*: they depend on the region i , not the time t . The only time dependence in the aggregated mean (10) is through the population offset m_{it} .

Estimation reuses the Gauss–Newton fixed point of the previous section unchanged, with the inner linear fit now the Kronecker-free space–time MCML (`SDALGCP2_ST`) rather than the spatial one. At iteration t the effective design row $\mathbf{D}_i^c(\boldsymbol{\beta}^{(t)})$ of (9) is replicated across the T times and the offset is

$$o_{it}^{(t)} = \log m_{it} + b_i(\boldsymbol{\beta}^{(t)}) - \mathbf{D}_i^c(\boldsymbol{\beta}^{(t)})^\top \boldsymbol{\beta}^{(t)}. \quad (11)$$

At a fixed point the same argument as in the spatial case shows $\boldsymbol{\beta}^*$ solves the likelihood equations of the nonlinear space–time model. When z is constant within each region the tilting collapses exactly to the region design, $\mathbf{D}_i^c = [1, z_i^\top]$ and $b_i = [1, z_i^\top] \boldsymbol{\beta}$, so the fit reduces to the ordinary space–time model — a check used in the package tests. Misaligned covariates are handled identically,

with the Berkson-corrected offset of the companion note (`SDALGCP2_ST(..., rasters = )` and `SDALGCP2_ST(..., covariates = )`).

Reference. Johnson, O., Diggle, P., Giorgi, E. (2019). A spatially discrete approximation to log-Gaussian Cox processes for aggregated disease count data. *Statistics in Medicine* 38, 4871–4887.